Jose Rizal University
College of Computer Studies and Engineering
Computer Engineering Department

CPE C403 – Practice Design 1
Design Project Proposal - Project Synthesis Form
1st semester, SY 2025 to 2026

Title of Design Project Proposal: NON-INVASIVE CHOLESTEROL SCREENING DEVICE USING HAND IMAGING AND ARTIFICIAL NEURAL NETWORKS	
Project Features/Description	
Hardware Features	
Raspberry Pi - Presents itself as the central controller which coordinates image capture, preprocessing, AI analysis and wireless communication.	
Camera Module - Records standardized hand images which are utilized to estimate cholesterol.	
White SMD LED Array (100-300 lumens) - It is a stable lighting source that ensures that the image in use is clear and that the light involved does not come from outside sources which will create distortion.	
OLED Display - reports real-time cholesterol classification status (Normal or High) and device status.	
Wi-Fi Module (IEEE 802.11) - This allows the transfer of data wirelessly between the device and the mobile application.	
Enclosure Hand Slot 3D-Printed - The design allows to place the hand of the user in the right position and avoid ambient light, which may distort readings.	
Regulated Power Supply - Powers all components safely and supports portable operation.	
Software Features	
Admin	
Admin dashboard to check the user activity and the performance of the entire system.	
User management feature for viewing, adding, editing or deleting user accounts.	
Report page that displays the cholesterol outcomes and filters information by age or level of risk of the user.	
Real time communication between device, app, and database through wireless data transfer.	
User	
A mobile application which shows the level of cholesterol results (Normal or High) in real-time.	
Login and authentication system to ensure better security of users health information	
User dashboard that displays the most recent cholesterol and general health summary.	
Health profile section that captures user information including his or her name, age and gender.	
History feature which stores and shows the past cholesterol test results with a time stamp.	
Trend monitoring to visualize the changes of cholesterol in several scans.	
Inbuilt health tips which give short-term suggestions to the user.	
Rolling back-up of data and accessing it remotely through the cloud.	
Easy to use interface that is easy to operate by all the age groups.	
Group Member	**Signature**
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2. Derilo, Ashley Shane B.	
3. Jontilano, John Jelord M.	
4. Villas, Rhynne Charz M.	
Technical Adviser: Dr. Richard Ramos	
Panel Approval	
1. Engr. Rosalina Estacio	
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Noted by:	
ROSALINA R. ESTACIO **Subject Adviser**	**Engr. Rosalina Estacio** Dept. Head, Computer Engineering
Engr. Liza R. Reyes **Dean, College of CSE**	

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